

1.

QUESTIONS 31 / 31 completed

Arithmetic

ARITHMETIC

Write a Java program that takes two integers as input from the user and calculates their sum, difference, product, and quotient using arithmetic operators.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class arithmetic{
3     public static void main(String[] args){
4         int a,b;
5         Scanner sc=new Scanner(System.in);
6         System.out.println("enter the two values");
7         a=sc.nextInt();
8         b=sc.nextInt();
9         System.out.println(a+b);
10        System.out.println(a-b);
11        System.out.println(a*b);
12        System.out.println(a/b);
13        System.out.println(a%b);
14
15    }
16 }
```

INPUT

2
3

OUTPUT

```
enter the two values
5
-1
6
0
2
```

2.

QUESTIONS 31 / 31 completed

MAX

MAX

Write a Java program to find the maximum of three numbers using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1
2 public class max{
3     public static void main(String[] args){
4         int a=10;
5         int b=20;
6         int c=30;
7         if(a>b&&a>c){
8             System.out.println("a is greater");
9         }
10        else if(b>a&&b>c){
11            System.out.println(" b is greater");
12        }
13        else{
14            System.out.println("c is greater");
15        }
16    }
17 }
```

INPUT

Your INPUT goes here!

```
13     else{
14         System.out.println("c is greater");
15     }
16 }
17 }
18 }
```

OUTPUT

c is greater

3.

QUESTIONS 31 / 31 completed

Leap

LEAP

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class learyear{
3     public static void main(String[] args){
4         int year;
5         Scanner sc = new Scanner(System.in);
6         System.out.println("enter the year");
7         year = sc.nextInt();
8         if((year % 400 == 0)|| (year % 4 == 0&&year%100 != 0)){
9             System.out.println("leap year");
10        }else {
11            System.out.println("not a leap year");
12        }
13    }
14 }
```

INPUT

2026

OUTPUT

enter the year
not a leap year

4.

QUESTIONS 31 / 31 completed

Fibonacci

FIBONACCI

Write a Java program to print the Fibonacci series up to a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class fibonacci{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number");
6         int n=sc.nextInt();
7         int a=0,b=1,c;
8         System.out.println("fibonacci series");
9         for(int i=1;i<=n;i++){
10             System.out.println(a+" ");
11             c=a+b;
12             a=b;
13             b=c;
14         }
15     }
16
17 }
```

INPUT

10

OUTPUT

```
enter the number
fibonacci series
0
1
1
2
3
5
8
13
```

5.

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Factorial{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("enter the number");
6         int n = sc.nextInt();
7         int fact = 1;
8         int i = 1;
9         while (i<=n){
10             fact = fact*i;
11             i++;
12         }
13         System.out.println("factorial of "+n+" "+fact);
14     }
15 }
```

INPUT

5

OUTPUT

```
enter the number
factorial of 5 120
```

6.

QUESTIONS 31 / 31 completed

RT-Pattern

RT-PATTERN

Write a Java program to print the following pattern using loops:

PROGRAM EDITOR

Java

Run

Save

```
1 public class StarPattern {
2     public static void main(String[] args) {
3         // Outer loop controls the number of rows (from 2 st
4         for (int i = 1; i <= 5; i++) {
5             // Inner loop prints the stars for the current r
6             for (int j = 1; j <= i; j++) {
7                 System.out.print("*");
8             }
9             // Move to the next line after completing a row
10            System.out.println();
11        }
12    }
13 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
**
***
****
*****
```

7.

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the number of elements: ");
9         int n = sc.nextInt();
10
11         int[] arr = new int[n];
12
13         System.out.println("Enter the sorted array elements:");
14
15         for (int i = 0; i < n; i++) {
16             arr[i] = sc.nextInt();
17         }
18
19         System.out.print("Enter the element to search: ");
20         int key = sc.nextInt();
```

5

10

20

30

40

50

50

ENGINEERING

JAVA PROGRAMMING

Search

SEARCH

Write a Java program to implement binary search on an array of integers using loops and conditional statements.

```
30         if (arr[mid] == key) {
31             found = 1;
32             System.out.println("Element found at index " + mid);
33             break;
34         }
35         else if (arr[mid] < key) {
36             low = mid + 1;
37         }
38         else {
39             high = mid - 1;
40         }
41     }
42
43     if (found == 0) {
44         System.out.println("Element not found.");
45     }
46
47     sc.close();
48 }
49 }
```

5

10

20

30

40

50

50

OUTPUT

```
Enter the number of elements: Enter the sorted array
elements:
Enter the element to search: Element not found.
```

8.

QUESTIONS 31 / 31 completed

GCD

GCD

Write a Java program to find the GCD (Greatest Common Divisor) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter first number: ");
9         int num1 = sc.nextInt();
10
11        System.out.print("Enter second number: ");
12        int num2 = sc.nextInt();
13
14        int gcd = 1;
15
16        for (int i = 1; i <= num1 && i <= num2; i++) {
17
```

INPUT

12
87

Common Divisor) of two numbers using loops and conditional statements.

```
14         int gcd = 1;
15
16         for (int i = 1; i <= num1 && i <= num2; i++) {
17
18             if (num1 % i == 0 && num2 % i == 0) {
19                 gcd = i;
20             }
21
22         }
23
24         System.out.println("GCD = " + gcd);
25
26         sc.close();
27     }
28 }
```

OUTPUT

```
Enter first number: Enter second number: GCD = 3
```


9.

QUESTIONS 31 / 31 completed

Prime

PRIME

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Prime{
3     public static void main(String[] str){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number");
6         int n = sc.nextInt();
7         boolean isprime=true;
8         if(n<=1){
9             is prime = false
10        }else{
11            for(int i = 2;i<=n/2;i++){
12                for (n%i==0){
13                    is prime = false;
14                    break;
15                }
16            }
17        }
```

INPUT

Your INPUT goes here!

10.

QUESTIONS 31 / 31 completed

Sort

SORT

Write a Java program to implement selection sort on an array of integers using loops and conditional statements.

on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i = 0;i<n;i++){
10             arr[i] = sc.nextInt();
11         }
12         for(int i =0;i<n-1;i++){
13             int minIndex = i;
14             for(int j=i+1;j<n;j++){
15                 if(arr[j]<arr[minIndex]){
16                     minIndex = j;
17                 }
18             }
19             int temp = arr[i];
20             arr[i] = arr[minIndex];
21             arr[minIndex] = temp;
22         }
23         System.out.println("sorted array:");
24         for(int i =0;i<n;i++){
25             System.out.print(arr[i] + " ");
26         }
27         sc.close();
28     }
29 }
```

INPUT

```
5
54
23
34
56
34
```

OUTPUT

```
enter the number of elements:enter the array elements:
sorted array:
23 34 34 54 56
```

11.

QUESTIONS 31 / 31 completed

Table

TABLE

Write a Java program to print the multiplication table of a given number using loops.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class multilplication{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number:");
6         int n = sc.nextInt();
7         for(int i=0;i<=10;i++){
8             System.out.println(n + " x " + i + " = " + (n*i));
9         }
10    }
11 }
```

INPUT

10

OUTPUT

```
enter the number:
10 x 0= 0
10 x 1= 10
10 x 2= 20
10 x 3= 30
10 x 4= 40
10 x 5= 50
10 x 6= 60
10 x 7= 70
10 x 8= 80
```

12.

QUESTIONS 31 / 31 completed

Factorial

FACTORIAL

2. Write a Java program to find the factorial of a given number using recursion.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class fact{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number");
6         int n = sc.nextInt();
7         int fact = 1;
8         for(int i =1;i<=n;i++){
9             fact = fact*i;
10        }
11        System.out.println("factorial:" + fact);
12    }
13 }
```

INPUT

5

OUTPUT

```
enter the number
factorial:120
```

13.

QUESTIONS 31 / 31 completed

LT-Pattern

LT-PATTERN

Write a Java program to print the following pattern using loops:*****

**
*

PROGRAM EDITOR

Java

Run

Save

```
1 public class LTPattern {  
2     public static void main(String[] args) {  
3         for (int i = 5; i >= 1; i--) {  
4             for (int j = 1; j <= i; j++) {  
5                 System.out.print("*");  
6             }  
7             System.out.println();  
8         }  
9     }  
10 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
*****  
****  
***  
**  
*
```

14.

QUESTIONS 31 / 31 completed

LCM

LCM

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter first number: ");
9         int num1 = sc.nextInt();
10
11        System.out.print("Enter second number: ");
12        int num2 = sc.nextInt();
13
14        int max;
15
16        if (num1 > num2) {
17            max = num1;
```

INPUT

12
87

Write a Java program to find the LCM (Least Common Multiple) of two numbers using loops and conditional statements.

```
18         } else {
19             max = num2;
20         }
21
22         while (true) {
23
24             if (max % num1 == 0 && max % num2 == 0) {
25                 System.out.println("LCM = " + max);
26                 break;
27             }
28
29             max++;
30         }
31
32         sc.close();
33     }
34 }
```

OUTPUT

```
Enter first number: Enter second number: LCM = 348
```

15.

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter a decimal number: ");
9         int decimal = sc.nextInt();
10
11         int binary = 0;
12         int place = 1;
13
14         while (decimal != 0) {
15
16             int remainder = decimal % 2;
17
```

INPUT

77

QUESTIONS 31 / 31 completed

D to B

D TO B

Write a Java program to convert a decimal number to binary using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
9         int decimal = sc.nextInt();
10
11         int binary = 0;
12         int place = 1;
13
14         while (decimal != 0) {
15
16             int remainder = decimal % 2;
17
18             binary = binary + (remainder * place);
19
20             place = place * 10;
21
22             decimal = decimal / 2;
23         }
24
25         System.out.println("Binary Number = " + binary);
26
```

INPUT

77

OUTPUT

```
Enter a decimal number: Binary Number = 1001101
```

16.

QUESTIONS 31 / 31 completed

B to D

B TO D

Write a Java program to convert a binary number to decimal using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter a binary number: ");
9         int binary = sc.nextInt();
10
11         int decimal = 0;
12         int power = 1;
13
14         while (binary != 0) {
15
16             int digit = binary % 10;
17
```

INPUT

11001

Write a Java program to convert a binary number to decimal using loops and conditional statements.

```
14         while (binary != 0) {
15
16             int digit = binary % 10;
17
18             decimal = decimal + (digit * power);
19
20             power = power * 2;
21
22             binary = binary / 10;
23         }
24
25         System.out.println("Decimal Number = " + decimal);
26
27         sc.close();
28     }
29 }
```

OUTPUT

```
Enter a binary number: Decimal Number = 25
```


17.

QUESTIONS 31 / 31 completed

Sum

SUM

Write a Java program to calculate the sum of the digits of a given number using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number:");
6         int n = sc.nextInt();
7         int sum =0;
8         while (n != 0){
9             int digit = n % 10;
10            sum=sum + digit;
11            n = n/10;
12        }
13        System.out.println("sum od digits="+sum);
14    }
15 }
```

INPUT

1234

OUTPUT

```
enter the number:
sum od digits=10
```

18.

Product

PRODUCT

Write a Java program to calculate the product of the digits of a given number using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the number:");
6         int num = sc.nextInt();
7         int product = 1;
8         while(num != 0){
9             int digit = num % 10;
10            product = product * digit;
11            num = num / 10;
12        }
13        System.out.println("product of digits" + product);
14    }
15 }
```

34

OUTPUT

```
enter the number:
product of digits12
```

19.

QUESTIONS 31 / 31 completed

Power

POWER

Write a Java program to find the power of a number using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the base: ");
9         int base = sc.nextInt();
10
11        System.out.print("Enter the exponent: ");
12        int exponent = sc.nextInt();
13
14        int power = 1;
15
16        if (exponent >= 0) {
17
```

INPUT

3
7

OUTPUT

```
Enter the base: Enter the exponent: Power = 2187
```

20.

QUESTIONS 31 / 31 completed

Area Circle

AREA CIRCLE

Write a Java program to calculate the area of a circle using the radius input by the user.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.println("enter the radius:");
6         double radius = sc.nextDouble();
7         double area = Math.PI*radius*radius;
8         System.out.println("area of circle:" + area);
9     }
10 }
```

INPUT

5.6

OUTPUT

```
enter the radius:
area of circle:98.52034561657591
```

21.

QUESTIONS 31 / 31 completed

Shpere ▾

SHPERE

Write a Java program to calculate the volume of a sphere using the radius input by the user.

PROGRAM EDITOR

Java ▾

Run Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter the radius: ");
9         double radius = sc.nextDouble();
10
11         double volume = (4.0 / 3.0) * Math.PI * radius * radius * radius;
12
13         System.out.println("Volume of Sphere = " + volume);
14
15         sc.close();
16     }
17 }
```

INPUT

4

OUTPUT

```
Enter the radius: Volume of Sphere = 268.082573106329
```

22.

QUESTIONS 31 / 31 completed

Palindrome

PALINDROME

Write a Java program to check whether a given string is a palindrome or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter a string: ");
9         String str = sc.nextLine();
10
11        String rev = "";
12
13        for (int i = str.length() - 1; i >= 0; i--) {
14            rev = rev + str.charAt(i);
15        }
16
17        if (str.equals(rev)) {
```

INPUT

MADAM

OUTPUT

```
Enter a string: Palindrome
```

23.

QUESTIONS 31 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter a number: ");
9         int num = sc.nextInt();
10
11         int original = num;
12         int sum = 0;
13
14         while (num != 0) {
15
16             int digit = num % 10;
17             sum = sum + (digit * digit * digit);
```

INPUT

1534

OUTPUT

```
Enter a number: 1534 is not an Armstrong Number.
```

24.

QUESTIONS 31 / 31 completed

RT-num.Pattern

**RT-
NUM.PATTERN**

Write a Java program to print the following pattern using loops:

1
22
333
4444
55555

PROGRAM EDITOR

Java

Run

Save

```
1 public class NumberPattern {  
2     public static void main(String[] args) {  
3         // Outer loop controls the number of rows  
4         for (int i = 1; i <= 5; i++) {  
5             // Inner loop controls the number of columns in  
6             for (int j = 1; j <= i; j++) {  
7                 System.out.print(i);  
8             }  
9             // Move to the next line after printing each row  
10            System.out.println();  
11        }  
12    }  
13 }
```

INPUT

Your INPUT goes here!

OUTPUT

```
1  
22  
333  
4444  
55555
```


25.

QUESTIONS 31 / 31 completed

Perfect

PERFECT

Write a Java program to check whether a given number is a perfect number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2
3 public class Main {
4     public static void main(String[] args) {
5
6         Scanner sc = new Scanner(System.in);
7
8         System.out.print("Enter a number: ");
9         int num = sc.nextInt();
10
11         int sum = 0;
12
13         for (int i = 1; i < num; i++) {
14
15             if (num % i == 0) {
16                 sum = sum + i;
17             }
18         }
19     }
20 }
```

INPUT

6

OUTPUT

```
Enter a number: 6 is a Perfect Number.
```

26.

Max Array

MAX ARRAY

Write a Java program to find the largest element in an array of integers using loops and conditional statements.

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i=0;i<n;i++){
10             arr[i] = sc.nextInt();
11         }
12         int largest = arr[0];
13         for(int i=1;i<n;i++){
14             if(arr[i]>largest){
15                 largest = arr[i];
16             }
17         }
18         System.out.println("largest elements:"+largest);
19     }
20 }
```

4
2
7
9
4

OUTPUT

```
enter the number of elements:enter the array
elements:
largest elements:9
```

27.

QUESTIONS 31 / 31 completed

2nd Max

2ND MAX

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i = 0; i < n; i++){
10             arr[i] = sc.nextInt();
11         }
12         int largest = Integer.MIN_VALUE;
13         int secondLargest = Integer.MIN_VALUE;
14         for(int i = 0; i < n; i++){
15             if(arr[i] > largest){
16                 secondLargest = largest;
17                 largest = arr[i];
```

INPUT

4
2
4
6
7

Write a Java program to find the second largest element in an array of integers using loops and conditional statements.

```
13     int secondLargest = Integer.MIN_VALUE;
14     for(int i = 0; i < n; i++){
15         if(arr[i] > largest){
16             secondLargest = largest;
17             largest = arr[i];
18         }
19         else if (arr[i] > secondLargest && arr[i] != largest)
20             secondLargest = arr[i];
21     }
22 }
23 if(secondLargest == Integer.MIN_VALUE){
24     System.out.println("Second largest element does not
25 }else{
26     System.out.println("second largest element =" + seco
27 }
28 }
29 }
```

6
7

OUTPUT

```
enter the number of elements:enter the array elements:
second largest element =6
```

28.

QUESTIONS 31 / 31 completed

Min

MIN

Write a Java program to find the smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter tge number os elements:");
6         int n= sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i = 0; i<n;i++){
10             arr[i] = sc.nextInt();
11         }
12         int smallest = arr[0];
13         for(int i = 1;i<n;i++){
14             if(arr[i] < smallest){
15                 smallest = arr[i];
16             }
17         }
18     }
19 }
```

INPUT

5
1
3
5
6
8

OUTPUT

```
enter tge number os elements:enter the array elements:
smallest element is1
```

29.

QUESTIONS 31 / 31 completed

2nd Min

2ND MIN

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java Run Save

```

1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i=0;i<n;i++){
10             arr[i] = sc.nextInt();
11         }
12         int smallest = Integer.MAX_VALUE;
13         int secondSmallest = Integer.MAX_VALUE;
14         for(int i=0;i<n;i++){
15             if(arr[i] < smallest){
16                 secondSmallest = smallest;
17                 smallest = arr[i];

```

INPUT

```

5
1
4
8
0
7

```

Write a Java program to find the second smallest element in an array of integers using loops and conditional statements.

```

14         for(int i=0;i<n;i++){
15             if(arr[i] < smallest){
16                 secondSmallest = smallest;
17                 smallest = arr[i];
18             }
19             else if(arr[i] < secondSmallest && arr[i] != smallest){
20                 secondSmallest = arr[i];
21             }
22         }
23         if(secondSmallest == Integer.MAX_VALUE){
24             System.out.println("second smallest element does not exist");
25         }else{
26             System.out.println("second smallest element = " + secondSmallest);
27         }
28     }
29 }

```

```

0
7

```

OUTPUT

```

enter the number of elements:enter the array elements:
second smallest element = 1

```

30.

QUESTIONS 31 / 31 completed

Sort-B

SORT-B

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the no of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i = 0; i<n; i++){
10             arr[i] = sc.nextInt();
11         }
12         for(int i = 0; i<n-1; i++){
13             for(int j=0; j<n-1-i; j++){
14                 if (arr[j] > arr[j+1]){
15                     int temp = arr[j];
16                     arr[j] = arr[j+1];
17                     arr[j + 1] = temp;
18                 }
19             }
20         }
21         System.out.println("sorted array:");
22         for(int i =0; i<n; i++){
23             System.out.println(arr[i] + " ");
24         }
25     }
26 }
```

INPUT

4
65
78
094
564

Write a Java program to implement bubble sort on an array of integers using loops and conditional statements.

```
10         arr[i] = sc.nextInt();
11     }
12     for(int i = 0; i<n-1; i++){
13         for(int j=0; j<n-1-i; j++){
14             if (arr[j] > arr[j+1]){
15                 int temp = arr[j];
16                 arr[j] = arr[j+1];
17                 arr[j + 1] = temp;
18             }
19         }
20     }
21     System.out.println("sorted array:");
22     for(int i =0; i<n; i++){
23         System.out.println(arr[i] + " ");
24     }
25 }
26 }
```

564

OUTPUT

```
enter the no of elements:enter the array elements:
sorted array:
65
78
94
564
```

31.

QUESTIONS 31 / 31 completed

Sort-I

SORT-I

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

PROGRAM EDITOR

Java

Run Save

```

1 import java.util.Scanner;
2 public class main{
3     public static void main(String[] args){
4         Scanner sc = new Scanner(System.in);
5         System.out.print("enter the number of elements:");
6         int n = sc.nextInt();
7         int[] arr = new int[n];
8         System.out.println("enter the array elements:");
9         for(int i =0; i<n;i++){
10            arr[i] = sc.nextInt();
11        }
12        for ( int i = 1; i<n; i++){
13            int key = arr[i];
14            int j = i-1;
15            while(j >=0&& arr[j]>key){
16                arr[j+1] = arr[j];
17                j--;

```

INPUT

5
34
78
7
87
0

Write a Java program to implement insertion sort on an array of integers using loops and conditional statements.

```

10         arr[i] = sc.nextInt();
11     }
12     for ( int i = 1; i<n; i++){
13         int key = arr[i];
14         int j = i-1;
15         while(j >=0&& arr[j]>key){
16             arr[j+1] = arr[j];
17             j--;
18         }
19         arr[j+1] = key;
20     }
21     System.out.println("sorted array:");
22     for(int i =0;i<n;i++){
23         System.out.print(arr[i]+" ");
24     }
25 }
26 }

```

87
0

OUTPUT

```

enter the number of elements:enter the array elements:
sorted array:
0 7 34 78 87

```